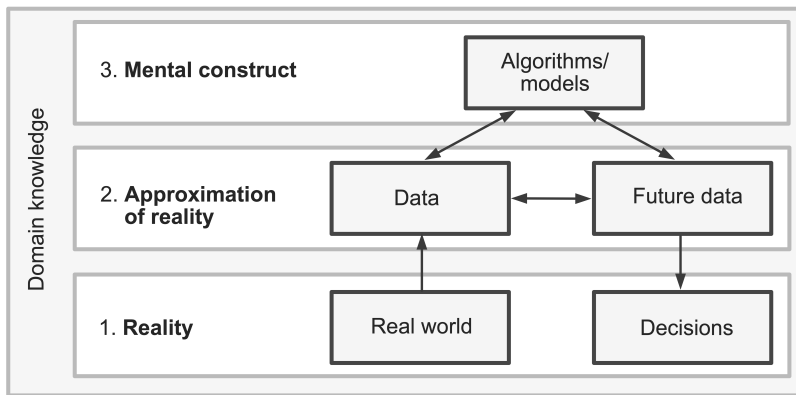


Assessing predictions

Ciaran Evans

Model assessment

Essentially, all models are wrong, but some are useful. (Box and Draper, 1987)



Today: Is our good at *predicting* the response?

Types of prediction

Regression:

Classification:

General principle: predictions from regression can be converted into classification, but not the other way around!

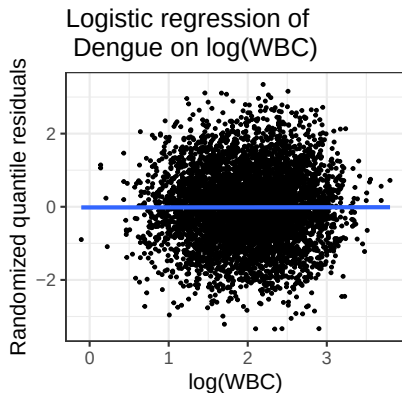
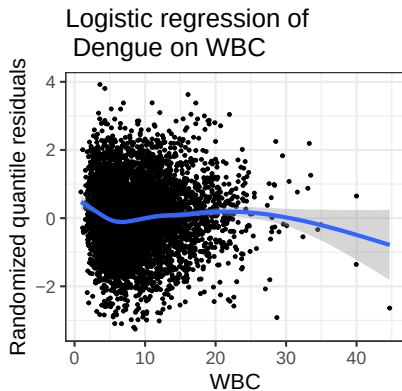
Assessing predicted probabilities

$$Y_i \sim \text{Bernoulli}(p_i)$$

$$\log \left(\frac{\hat{\pi}_i}{1 - \hat{\pi}_i} \right) = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_k X_{ik}$$

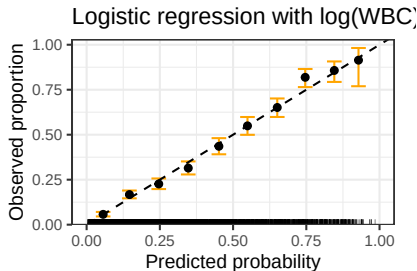
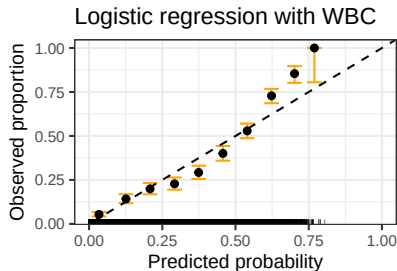
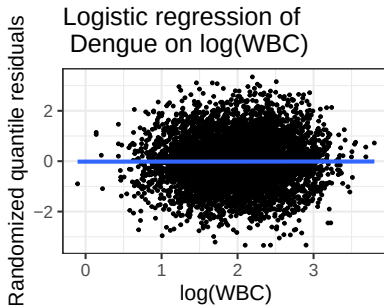
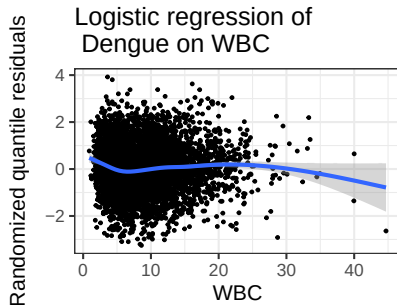
Question: How do we tell if $\hat{\pi}_i$ is a “good” estimator of π_i ?

Assessing predicted probabilities

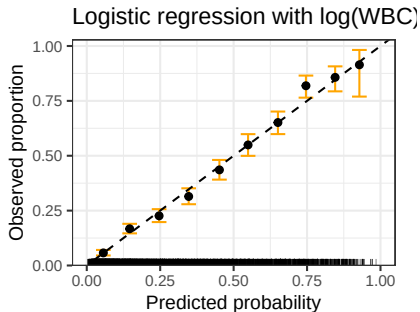
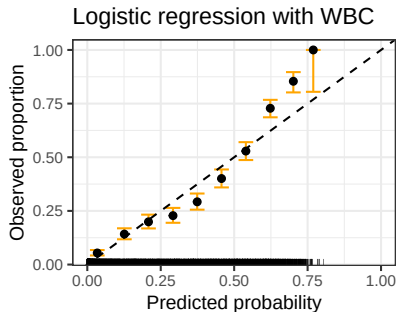


Question: Does using WBC vs. $\log(\text{WBC})$ make a difference for the quality of our predictions?

Calibration plot

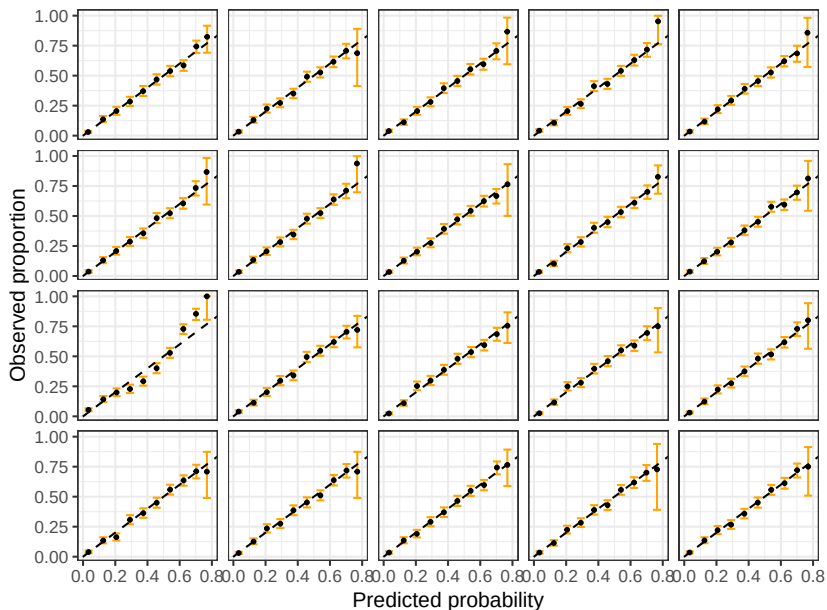


Calibration plot



Question: How do I tell if any unusual features in the calibration plot are just due to random chance?

Parametric bootstrap + visual inference

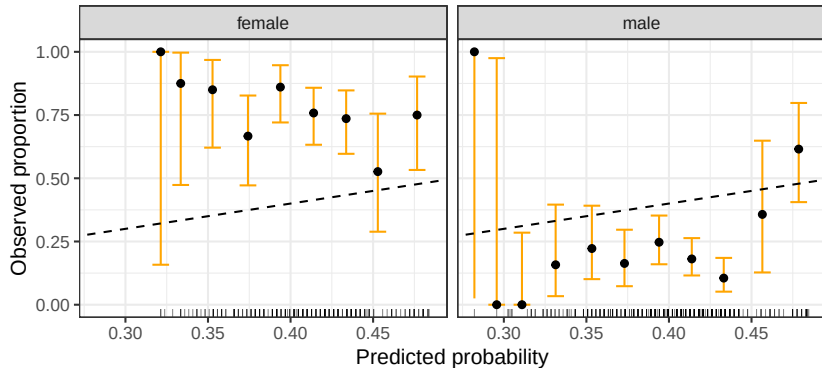


Class activity

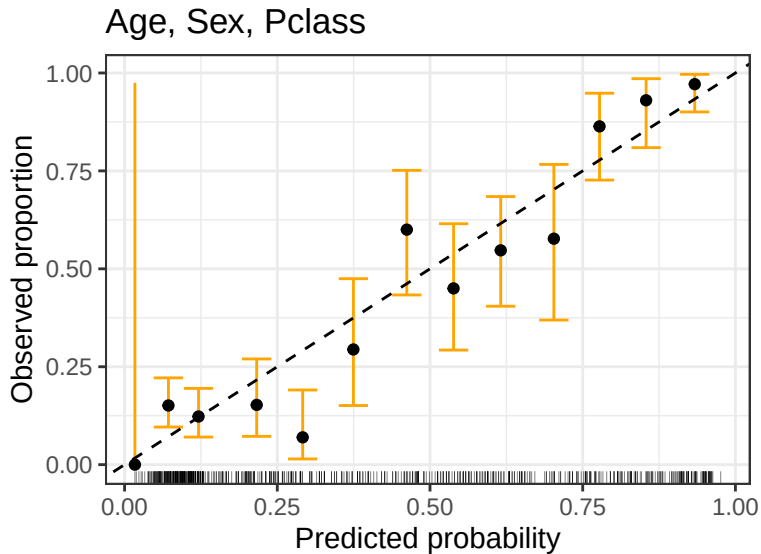
- ▶ Work on the class activity (course website) with a neighbor
- ▶ At the end of class, submit your rendered HTML on Canvas

Class activity

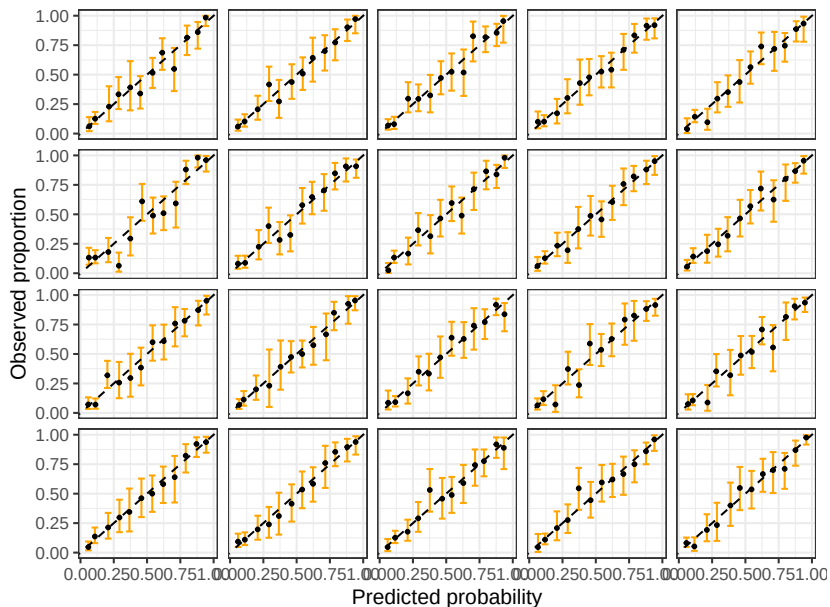
Age only, plotted by Sex



Class activity



Visual inference



Activity

Age, Sex, Pclass, plotted by Sex

